

Where Language Lives: Investigating Code-Switching Across Modalities in Small Vision-Language Models

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Abstract

Code-switching is the alternation between two languages within a single utterance. It happens constantly in everyday speech and text, and increasingly in images too, such as a screenshot of a Hinglish tweet or a photo of a handwritten note. In vision-language models (VLMs), we do not know whether a model reads that text using the same attention heads it would use for plain text tokens, or whether reading language through pixels calls on a different set of heads entirely. We are not aware of prior work that tests this directly for code-switched content, comparing the same sentence as text and as a rendered image at head-level granularity. We test this with a three-condition design (A = text, B = image+query, C = image-only) on real Hindi-English and Spanish-English code-switched sentences, rendered as images and evaluated across four open, sub-3B VLMs spanning three decoder lineages. Using gradient-based head-importance scoring (LAHIS), we find that two of the four models keep text and image pathways separate, one nearly merges them for Hindi, and one shows no strong pattern either way. All four vision encoders can already tell Latin from Devanagari script by their first layer, though how well varies widely across models (0.63–0.99 peak balanced accuracy). Which heads a model treats as language-important tracks decoder lineage more closely than vision encoder or language competence. The overlap falls as lineage distance grows, from 18 of 20 shared heads for a same-lineage pair down to zero for the most distant pair. With only four models, this is an observation rather than an independently replicated finding. Zero-ablating these heads shifts output script well above a random-head control in every model tested, which indicates these heads causally affect the output script rather than merely correlating with it. We release the pipeline, the code-switched corpus, and the per-head results.

1 Introduction

A phone screenshot of a Hinglish tweet, a photographed handwritten note, and a chat screenshot pasted into a vision-language assistant all put the same problem to a VLM. It has to identify and route two languages within a single image, without the token-level segmentation a text-only language model gets for free.

A VLM has two main parts. A vision encoder turns an image into a sequence of tokens, much as a tokenizer turns text into a sequence of tokens. A language decoder then reads whichever tokens it is given, whether they came from text or from the vision encoder, and generates output text. We investigate whether its attention heads behave the same way regardless of which path the tokens took to reach it, text or image.

Prior mechanistic work on multilingual routing has focused almost exclusively on the text-only case. [Liu et al. \(2026\)](#) introduce Language Attention Head Importance Scores (LAHIS) and show that a small set of attention heads in multilingual LLMs matters far more than the rest for language identification, discoverable via a single forward-backward pass with a differentiable head mask. [Nie et al. \(2025\)](#) trace language-confusion failures in English-centric LLMs to specific “confusion point” positions and final-layer transitions. [Wendler et al. \(2024\)](#) show that intermediate Llama-2 representations are closer to an English pivot than to the input or output language.

On the vision-language side, Ye et al. (2025) identify cross-lingual attention misalignment in VLMs and intervene on it to reduce multilingual object hallucination, and Baek et al. (2025) characterize a broad, low-similarity population of “OCR heads” responsible for reading text embedded in images, but restrict their study to English text on English-centric models.

Whether a VLM’s OCR pathway and its text pathway rely on the same circuitry matters for how far existing interpretability work already takes us. If a VLM converges on the same language-routing heads regardless of modality, everything known about language heads from text-only interpretability work should transfer directly to the visual case.

If the pathways diverge, and if which heads matter depends on decoder pretraining rather than on vision-language training, then visual language identification is doing separate work. That work is invisible to text-only interpretability, and its findings cannot be assumed to generalize across model families. We study three questions.

- **RQ1 (Modality).** When the same code-switched content reaches a VLM as text tokens and as a rendered image, does the model route it through the same attention heads, or through separate ones?
- **RQ2 (Predictor).** What predicts *which* heads a VLM treats as language-important, its vision encoder, its language competence, or its decoder lineage?
- **RQ3 (Causality).** Are the identified heads causally responsible for the model’s output script, or only correlated with it?

We study these questions with a three-condition design (Section 5), applied to real, human-produced Hindi-English and Spanish-English code-switched sentences and evaluated on four open VLMs spanning three decoder lineages. Our findings map to the three questions.

- **Availability.** Vision encoders linearly separate script identity (Latin vs. Devanagari) from the earliest layers in all four models, with fidelity ranging from near-chance (0.61–0.63 balanced accuracy, never improving with depth) to near-ceiling (0.99 by the eighth of twenty-four layers). Two models sharing the SigLIP-SO400M encoder family land at opposite extremes. The script signal is therefore available to the decoder, so any divergence in routing is not for lack of signal.
- **RQ1.** Text and image pathways diverge. Two of the four models keep them substantially separate, one nearly unifies them for a single tested language, and one shows no strong pattern. A strong encoder script signal is not sufficient for unified routing.
- **RQ2.** Head-coordinate overlap tracks decoder lineage more closely than vision encoder or language competence, and is graded by decoder similarity. The two models sharing a decoder family share 18 of 20 top head coordinates, whereas cross-lineage pairs share strictly fewer (a Gemma-based model 8–9 with the shared-lineage pair, a Llama-based model only 1–2, and the two most distant lineages zero), with all shared heads confined to the earliest layers. The ordering holds even for English, where language competence is closely matched. With four models across three lineages this is suggestive rather than an independently replicated cause.
- **RQ3.** Zero-ablation confirms, in all four models, that the heads identified by gradient importance are functionally responsible for output script, with effect sizes from 1.6 up to $8.2\times$ a random-head control (Table 3). In two of the four models, mean-ablation of the English-selective heads is *more* disruptive than zero-ablation, the opposite of the pattern in the other two.

2 Related Work

Language-specific heads and neurons in text LLMs. Beyond LAHIS, Tang et al. (2024) identify language-specific neurons via parallel-corpus activation contrasts and show that deactivating them degrades

only the corresponding language. [Wendler et al. \(2024\)](#) probe intermediate representations directly for an English-pivot signature. This line of work establishes that language routing in LLMs is sparse, identifiable, and at least partially causal.

It is, however, entirely text-only. None of it asks what happens when the same content is presented as an image instead of a token sequence, and none of it asks whether the *identity* of important heads is a property of the decoder family or specific to one training run.

Code-switching circuits. [Nie et al. \(2025\)](#) is the closest prior work to a mechanistic study of multilingual routing failure, localizing language-confusion errors to specific token positions and late-layer transitions in text-only models. We are not aware of any prior mechanistic study of code-switching specifically (as opposed to single-language confusion), and none that touches the visual modality.

Language heads in VLMs. [Ye et al. \(2025\)](#) (CLAIM) is the closest prior work to ours. It identifies cross-lingual attention misalignment in VLMs and intervenes on it. The critical differences are that CLAIM studies monolingual queries in different languages rather than code-switched content, and it does not perform the decoder-extraction text-vs-image comparison central to our design. CLAIM never asks whether a VLM’s *text* decoder and its *image-reading* decoder agree on which heads matter, because the text condition is never isolated from the vision pathway.

[Baek et al. \(2025\)](#) characterize OCR heads in VLMs but restrict their study to English text, leaving open whether script-mixed or language-mixed visual text is routed any differently.

Gap. To our knowledge, no prior work (a) evaluates code-switching in the visual input channel of a VLM, (b) compares the same content as text versus image at head-level granularity with the confounding query removed, (c) asks whether script identity (orthographic system) or language identity is the more fundamental unit of visual language routing, or (d) tests whether cross-modal head identity is predicted by decoder lineage using a controlled same-lineage-vs-different-lineage comparison. We investigate all four questions in this paper.

3 Data

All code-switched text is real and human-produced, with gold token-level language labels. We avoid synthetic code-switching, such as naive word substitution, since it risks sentences no bilingual speaker would actually write.

Hindi-English data comes from the Devanagari-script Twitter portion of the GLUECoS part-of-speech corpus ([Khanuja et al., 2020](#)), and Spanish-English data comes from the Bangor Miami corpus of transcribed bilingual speech ([Deuchar et al., 2013](#)). Monolingual baselines in English, Spanish, and Hindi come from FLORES-200 parallel triples.

Cleaning and final counts. We clean the raw social-media and transcribed-speech text with Penn Treebank detokenization, script- and noise-consistency filters, a two-stage fluency filter (heuristic checks, then Qwen2.5-0.5B perplexity, dropping the worst quartile per language pair), and a content filter for profanity and political material judged by Claude against a fixed rubric, keeping every dropped record’s drop reason for auditability.

After filtering, the corpus contains 933 code-switched sentences (806 Hindi-English, 127 Spanish-English), 118 natural held-out Hindi-English sentences disjoint from that set, 750 monolingual sentences (250 per language), and 100 Hindi-in-Latin-script transliteration controls, which separate script identity from language identity since this content is Hindi written in the Latin alphabet.

Rendering. Every sentence exists as both raw text and a rendered image, black text on a white canvas, with per-token bounding boxes so patch-level attention can later be mapped back to source-language spans. Code-switched sentences render each token in a script-appropriate font, Noto Sans for Latin and Noto Sans

Devanagari for Devanagari, so script boundaries are visually real rather than assumed. Appendix E shows sample rendered stimuli (Figure 5).

4 Models

We evaluate four open VLMs, chosen to differ in vision encoder, decoder lineage, and image-token compression scheme while remaining small enough (1.7–3B total or decoder parameters) to run full LAHIS gradient attribution (forward and backward passes with a per-head soft mask) on the hardware available to us, a single NVIDIA T4 GPU with 16GB of VRAM. Model size was chosen to fit this budget, not for any property of the models themselves.

- **Qwen2-VL-2B-Instruct** (Wang et al., 2024) pairs a custom 2D-RoPE vision transformer (0.67B) with a 2×2 patch-merger and a Qwen2-1.5B decoder (28 layers, 12 attention heads).
- **InternVL3-2B** (Chen et al., 2024) pairs InternViT-300M (v2.5), with pixel-shuffle compression, and a Qwen2.5-1.5B decoder (28 layers, 12 attention heads).
- **SmolVLM-Instruct** (Marafioti et al., 2024) pairs a SigLIP-SO400M (Zhai et al., 2023) vision encoder and a fixed-scale-factor pixel-shuffle connector with a SmolLM2-1.7B decoder (24 layers, 32 attention heads).
- **PaliGemma-3B (mix-448)** (Beyer et al., 2024) pairs a SigLIP-SO400M vision encoder at 448px and a linear projector that keeps all 1024 image tokens (no merging or pooling) with a Gemma-2B decoder (18 layers, 8 attention heads, multi-query attention). It is a *prefix-LM*, so image and prompt tokens receive bidirectional attention and only the suffix is trained causally. We therefore score its decoder in that native suffix regime, taking the loss on the text tokens of `[BOS] \n text`, for both the competence gate and the LAHIS A condition.

Qwen2-VL-2B and InternVL3-2B share a decoder *lineage* (Qwen2 and Qwen2.5, both from the same model family and pretraining pipeline generation); SmolVLM’s SmolLM2 decoder is Llama-architecture and PaliGemma’s is Gemma, each from an unrelated pretraining lineage. This gives us a same-lineage pair plus two further decoder lineages, spanning three lineages in four models, which we use directly in Section 6.

Language competence gate. Before running any language-routing analysis, we verify that each extracted decoder is competent in each language, so that head-importance scores do not measure noise. We compute decoder perplexity on monolingual FLORES text (Table 1), using a per-token threshold of 100. All four models pass, though PaliGemma-3B’s per-token Spanish (114.7) slightly exceeds it.

Per-token perplexity must not be compared across languages, because tokenizer fertility differs by 3.4 to 3.6 times between English (about 0.23 tokens per character) and Devanagari Hindi (about 0.78). This mechanically deflates Hindi’s per-token perplexity regardless of competence. We therefore normalize by characters, using $\text{ppl}_c = \exp(\text{NLL}/n_{\text{chars}})$. On this scale Hindi is the hardest language for every model, and all cross-language competence statements in this paper use the character-normalized values.

5 Method

Figure 1 gives an overview of this pipeline before we describe each part in detail.

5.1 Three conditions

For each language and each sentence, we evaluate three conditions:

- A (text)** The extracted decoder processes the raw sentence text. This decoder is the language model sub-module taken *from inside* the VLM after its own vision-language instruction tuning, not a separately loaded checkpoint. The vision encoder never runs.

Table 1: Decoder language competence on monolingual FLORES text. Per-token perplexity is deflated for Hindi by tokenizer fertility and is not comparable across languages. The character-normalized values are comparable and show Hindi is hardest for every model, with all models in range. PaliGemma-3B is scored in its native prefix-LM regime.

Model	Per-token ppl.			Char.-normalized ppl.		
	en	es	hi	en	es	hi
Qwen2-VL-2B	40.9	26.2	5.9	2.06	2.23	4.46
InternVL3-2B	38.4	28.0	6.7	2.04	2.28	4.87
SmolVLM	83.8	49.5	7.4	2.23	2.98	7.64
PaliGemma-3B	91.3	114.7	69.4	2.48	2.75	4.48

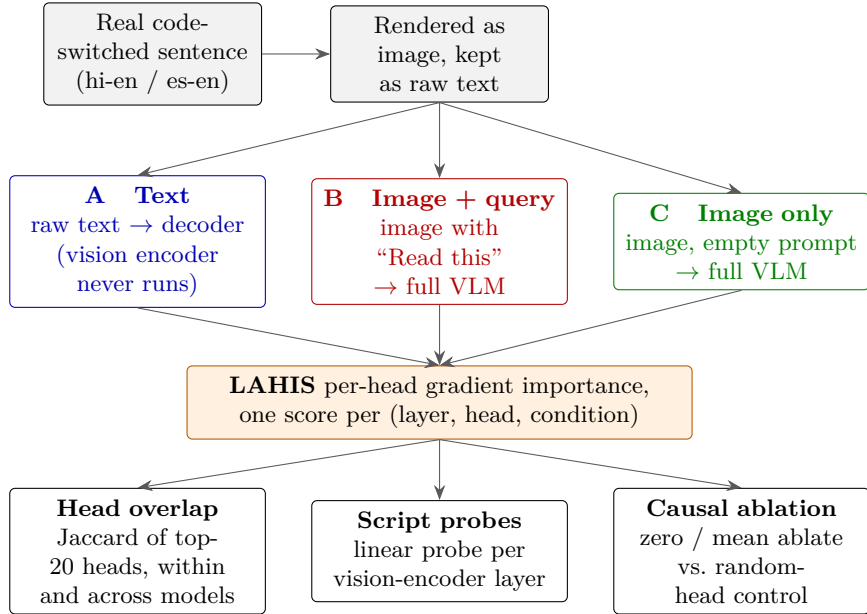


Figure 1: Overview of the pipeline. Each code-switched sentence is evaluated in three conditions (A, B, C), scored with LAHIS gradient attribution, and the resulting per-head importance scores feed three analyses, namely head overlap, script probes, and causal ablation.

B (image+query) The full VLM processes the rendered image plus a fixed textual instruction (“Read this:”), with the loss computed on the appended target sentence.

C (image-only) Identical to B but with an empty instruction. It uses the same processor chat template as B (image block, user and assistant role tokens), with the user’s text field left empty, so it adds no instruction-language tokens beyond the role tokens that B and C already share.

Using the decoder *extracted from* the VLM for condition A, rather than the original pre-finetuning checkpoint, is essential. It ensures that A vs. C isolates modality alone, without confounding modality with whether the weights have seen vision-instruction tuning at all.

5.2 Head importance (LAHIS)

Following Liu et al. (2026), we install a differentiable soft mask $M \in \mathbb{R}^{L \times H}$ (initialized to all ones) that multiplies each attention layer’s per-head output via a forward pre-hook on the output projection. This works because the projection’s *input* is the concatenation of per-head outputs and can therefore be sliced

exactly per head (unlike hooking the post-projection output, which mixes heads). We compute the language-modeling loss and take $|\partial \text{loss} / \partial M|$ as the importance score for each head, averaged over 100 sentences per condition per language. This requires only one forward and one backward pass per sentence.

We visualize these per-head importance scores as layer $\times$ head maps in Appendix D, for Hindi (Figure 3) and Spanish (Figure 4).

5.3 Head overlap

For each model and language, we compute the top-20 heads by importance under each condition and report pairwise Jaccard overlap ($|A \cap B| / |A \cup B|$) between conditions. We additionally compute a 10,000-sample permutation baseline (randomly-drawn 20-head sets from the same layer $\times$ head grid, computed separately per model since the grids differ in size) to establish chance-level overlap, and sweep $K \in \{5, 10, 20, 40, 80\}$ to check that our headline results are not an artifact of the specific top-20 cutoff.

To compare *across* models, we additionally record the raw (layer, head) coordinates in each model’s top-20 text-condition set and compute pairwise coordinate overlap between every pair of models.

5.4 Script probes

To ask whether script identity is already linearly decodable inside the vision encoder, before any information reaches the decoder, we train balanced-accuracy logistic-regression probes (5-fold cross-validated) at every vision-encoder layer, predicting each patch’s script (Latin, Devanagari, or background) from its frozen hidden state. Patch-to-script labels come from the token bounding boxes recorded at render time, mapped onto the actual grid of image tokens the decoder sees.

This grid differs from the raw patch grid whenever a compression module, such as a patch merger or pixel-shuffle layer, runs between encoder and decoder. Background patches are labeled only to be excluded. Every patch we actually fit or score the probe on is Latin or Devanagari, so the reported classifier is binary and chance is 0.5, not 1/3.

5.5 Causal ablation

Head importance from gradient attribution is correlational. A head can receive a large gradient without being causally responsible for the model’s output. We test causality directly by zero-ablating the top-20 heads found under condition A for English and for Hindi separately, via the same per-head output-projection hook, now clamping the targeted heads’ output to zero, and generating text from each of 100 held-out Hindi-English code-switched images. Generation is greedy and deterministic (no sampling, 64-token budget), so ablation variants differ only in which heads are masked, not in decoding randomness.

We record whether the output script changes from the un-ablated baseline. We restrict this test to Hindi-English images because the outcome measure is a script change, and script is only informative when the two code-switched languages do not share one. Spanish and English are both Latin-script, so a script-flip metric cannot register a language change for that pair. We handle Spanish-English separately with a language-identification metric (Table 4).

We compare against two controls. The first is a random 20-head set, drawn from heads *outside* the top-20 English and Hindi sets, to test whether *any* 20-head ablation degrades output this much. The second is a mean-ablation variant, which replaces each head’s output with its mean taken over token positions within that same image’s baseline forward pass (per-sentence, per-head, not corpus-wide) rather than zero, to separate language-selective suppression from generic degradation.

6 Results

6.1 Availability: script identity is decodable early in the vision encoder

Script-probe balanced accuracy exceeds the 50% chance level even at layer 0 for all four encoders, confirming script identity is present in the earliest patch embeddings before any contextualization across the image.

The four encoders differ substantially in how well. InternVL3-2B’s InternViT reaches 0.990 balanced accuracy by layer 8 of 24 and stays above 0.98 through its final layer, near-perfect linear separability. PaliGemma-3B’s SigLIP-SO400M encoder is nearly as strong, 0.982 at layer 12 of 27, climbing from an already-high 0.818 at layer 0. Qwen2-VL’s custom encoder peaks lower, 0.811 at layer 12 of 32.

SmolVLM’s SigLIP-SO400M encoder is the weakest by a wide margin and, unlike the other three, never improves substantially with depth. It peaks at only 0.630 (layer 8 of 27) and *declines* to 0.605 by its final layer, essentially flat, near-chance-plus performance throughout the network.

PaliGemma-3B and SmolVLM share the same vision-encoder family (SigLIP-SO400M), yet sit at opposite extremes, near-perfect script separability (0.982) against barely-above-chance (0.630). The difference is how the encoder is deployed. PaliGemma runs SigLIP at 448px and keeps all 1024 patch tokens, whereas SmolVLM feeds it lower-resolution, pooled tokens. So even within one encoder family the script signal reflects the resolution and token budget, not the architecture alone. All four models are given the same rendered images with the same fonts, so this spread comes from the encoders rather than the stimuli. Figure 2 shows the full per-layer trajectory behind these numbers.

These probes track rendered script rather than language content. We verify this across all four models using the 100 Hindi-in-Latin-script transliteration controls (Section 3). These images are Hindi *words* written in Latin *glyphs*. We apply each model’s already-trained, peak-layer Latin-versus-Devanagari probe to them, and the probe calls them Latin at a rate matching its own code-switched-data accuracy (InternVL3-2B 99.7%, PaliGemma-3B 99.7%, Qwen2-VL-2B 97.3%, SmolVLM 84.0%, against 98.8%, 98.2%, 80.2%, and 62.4% balanced accuracy on real script respectively). The probes are reading orthography, not guessing at the underlying language from lexical cues leaking into the patch representation.

6.2 RQ1: text and image pathways diverge

Table 2 reports pairwise Jaccard overlap of top-20 heads across the three conditions, per model and language, against a permutation-derived chance level (95th percentile over 10,000 random 20-head draws; SmolVLM’s larger 24×32 head grid gives it a lower chance level, 0.053, than the other two models’ 28×12 grid, 0.081).

Across four models, a majority pattern emerges. **Two of the four, Qwen2-VL-2B and SmolVLM, keep text and image pathways substantially separate.** $A \cap C$ stays low across every language tested (0.05–0.21), while the two image conditions cluster tightly together ($B \cap C$ 0.43–0.91, reaching its highest values for SmolVLM). For both models, once an image is present, a consistent head set is recruited largely independent of whether a text instruction accompanies it, and that set is substantially different from the text-reading head set.

InternVL3-2B is the outlier, and only for Hindi. Its $A \cap C$ for Hindi reaches 0.667 and $A \cap B$ reaches 0.905, close to unified routing between reading Hindi as tokens and reading the identical sentence as pixels. Its English and Spanish overlaps are lower and pattern more like the other models, though still generally higher. This unification is not the default across the four models we test, nor a byproduct of superior Hindi competence, since InternVL3-2B’s character-normalized Hindi perplexity (4.87) is no better than Qwen2-VL-2B’s (4.46), which does not unify. It appears specific to InternVL3-2B’s combination of vision encoder and decoder, on this particular language.

PaliGemma-3B shows no strong condition structure either way. Its overlaps are compressed into a narrow 0.21–0.33 band that sits only modestly above its elevated chance level (0.143, a consequence of its small 18×8 head grid), and its image-image overlap ($B \cap C$ 0.25–0.33) is not distinctly higher than its text-image overlaps. It therefore shows neither the clean image-pathway clustering of Qwen2-VL-2B and SmolVLM nor the InternVL3-2B-style merge. We read it as a weak, chance-dominated signal rather than

Table 2: Jaccard overlap of top-20 heads across conditions. A = text, B = image+query, C = image-only. Chance (p_{95}): 0.081 (Qwen2-VL-2B, InternVL3-2B), 0.053 (SmolVLM), 0.143 (PaliGemma-3B, whose small 18×8 head grid raises its chance level).

Model	Lang	$A \cap B$	$A \cap C$	$B \cap C$
Qwen2-VL-2B	en	0.143	0.081	0.538
	es	0.081	0.053	0.538
	hi	0.111	0.176	0.429
InternVL3-2B	en	0.333	0.176	0.176
	es	0.538	0.250	0.212
	hi	0.905	0.667	0.667
SmolVLM	en	0.176	0.143	0.667
	es	0.111	0.111	0.905
	hi	0.250	0.212	0.905
PaliGemma-3B	en	0.250	0.333	0.333
	es	0.212	0.290	0.250
	hi	0.250	0.290	0.333

evidence for either regime. This is a reminder that the condition-overlap measure loses resolution when the head grid is small.

This is not an artifact of the $K=20$ cutoff. Sweeping $K \in \{5, 10, 20, 40, 80\}$ (full results in Appendix B), text-image overlap ($A \cap B$, $A \cap C$) stays below image-image overlap ($B \cap C$) for Qwen2-VL-2B and SmolVLM at every K we tested, and InternVL3-2B’s three Hindi conditions stay close together (0.58–1.0) at every K , the same qualitative pattern Table 2 shows at $K=20$.

6.3 RQ2: head-coordinate overlap tracks decoder lineage

Within each model, most of the top-20 text-condition heads are shared across all three languages. Qwen2-VL-2B shares 18 of 20, InternVL3-2B 14 of 20, SmolVLM 14 of 20, and PaliGemma-3B 15 of 20. Most of the heads that this measure calls “language-important” are therefore language-*general* rather than language-*specific*, in every model tested.

The more important result is *across* models, and it is graded by decoder similarity. Qwen2-VL-2B and InternVL3-2B, which have different vision encoders but the same decoder lineage (Qwen2 and Qwen2.5), share 18 of their 20 top text-condition head coordinates. This is a raw intersection count, not a Jaccard fraction; the equivalent Jaccard is $18/22 \approx 0.818$. Every cross-lineage pair shares strictly fewer, and all shared coordinates lie in the first two layers, forming an early, near-universal core.

PaliGemma-3B (Gemma) shares 8 coordinates with Qwen2-VL-2B and 9 with InternVL3-2B. SmolVLM (Llama-family SmolLM2) shares only 2 with Qwen2-VL-2B and 1 with InternVL3-2B. The two most distant lineages, SmolVLM and PaliGemma-3B, share *zero*. The picture is therefore not a clean binary in which any different-lineage model shares almost nothing. The Gemma model shares a moderate part of the early core with the Qwen pair. But the same-lineage pair is the clear maximum (18), and sharing falls off with lineage distance.

We can address two candidate confounds directly. Competence does not explain the ordering. SmolVLM is the weakest model on Hindi by character-normalized perplexity (7.64 against 4.46–4.87) yet shares the fewest heads, while PaliGemma, competent on every language, shares an intermediate amount. The ordering also holds language by language. Even for English, where SmolVLM’s competence is within 10% of the Qwen pair (char-ppl 2.23 against 2.04–2.06), it still shares only 2 and 1 English top-20 coordinates against the Qwen pair’s 13. A competence account would predict sharing to track the competence gap, and it does not.

Table 3: % of 100 held-out hi-en code-switched generations whose output script flips from the un-ablated baseline, by ablation variant (top language-selective heads for the named language). The four columns are Qwen2-VL-2B, InternVL3-2B, SmolVLM, and PaliGemma-3B. PaliGemma’s random-head control is elevated (31%) because its small 18×8 head grid makes 20 heads a large fraction of the total; its effect sizes should be read against that baseline (hi-zero = $3.1 \times$ control).

Variant	Qwen	Intern	Smol	PaliG
Zero-ablate en	53%	25%	37%	18%
Zero-ablate hi	74%	14%	37%	97%
Mean-ablate en	33%	20%	52%	61%
Mean-ablate hi	29%	24%	18%	30%
Random-20 (control)	9%	9%	9%	31%

Vision encoder does not track it either. PaliGemma and SmolVLM use the same SigLIP-SO400M family but share zero heads, while PaliGemma and the InternViT and Qwen encoders differ yet share 8 to 9. Decoder lineage is therefore the best-supported explanation among the factors we can rule in or out here, though it is still not an independently replicated cause. Section 8 details the remaining covariates.

Extended to three lineages, the pattern is consistent with the LAHIS finding that language heads reflect the base decoder more than downstream training. We observe this for the heads a VLM recruits to read language *out of an image*, which the decoder alone was never trained to do.

6.4 RQ3: ablation confirms the heads are causally responsible

Zero-ablating each model’s top-20 language-selective heads flips output script well above its random-head control on 100 held-out Hindi-English code-switched images (Table 3). This confirms that the heads identified by gradient attribution are causally responsible for output script rather than just correlated with language processing, in every model tested. The control is 9% for the three causal-decoder models. For PaliGemma it rises to 31%, because its small 18×8 head grid makes a random 20-head set a large fraction of all heads. Even so, its Hindi zero-ablation flips 97% of generations, which is $3.1 \times$ that control and the strongest Hindi effect in the roster.

Bootstrap 95% confidence intervals (10,000 resamples over the 100 images) confirm that these zero-ablation effects are separable from the random-head control with no interval overlap. For example, PaliGemma-3B’s hi-zero is 97% [93,100] against its 31% [22,40] control, and Qwen2-VL-2B’s hi-zero is 74% [65,82] against 9% [4,15].

SmolVLM’s baseline is also more likely to produce Devanagari script unprompted than the other two models (37/100 against 5/100 and 11/100). We previously attributed this to stronger relative Hindi competence, but under the character-normalized metric SmolVLM is the weakest model on Hindi (Section 4). This behavioral difference is therefore a property of its decoding, not a competence effect, and we leave it unexplained.

For Qwen2-VL-2B and InternVL3-2B, mean-ablation is consistently *gentler* than zero-ablation, as expected if part of the zero-ablation effect is generic degradation rather than purely language-selective suppression. **SmolVLM and PaliGemma break this pattern for English.** Mean-ablating SmolVLM’s top-English heads (52% [42,62]) is nominally *more* disruptive than zero-ablating them (37% [28,47]), though these bootstrap intervals overlap, so the SmolVLM reversal is only suggestive; PaliGemma shows the same reversal far more sharply and cleanly (mean 61% [51,70] vs. zero 18% [11,26], no interval overlap).

One possibility is that these models’ top-English heads’ *mean* activation on this stimulus set is itself informative for English continuation, so replacing a head’s output with that mean pushes the output toward English rather than only removing signal the way zeroing does.

For PaliGemma the reading is complicated by a ceiling effect. Its baseline is 85% Latin-script, so zero-ablation of English heads has little room to flip script, which lowers the zero-ablation figure. This reversal

Table 4: Spanish-English ablation. Percent of 100 es-en generations whose dominant language (`langid`, es vs. en) flips from the un-ablated baseline, zero-ablating the top-20 es-selective or en-selective heads, against the random-head control. The effect clears the control in every model, but the es/en asymmetry is far weaker than the hi/en asymmetry in Table 3, and the control is raised by language-ID noise on short generations.

Model	es-heads	en-heads	Random-20
Qwen2-VL-2B	17%	14%	8%
InternVL3-2B	48%	53%	33%
SmolVLM	39%	51%	8%
PaliGemma-3B	85%	82%	19%

is only suggestive in SmolVLM, but it appears with non-overlapping intervals in a second, architecturally unrelated model. That makes it more likely to be a real property of mean-ablation on visual English routing than an artifact. We report it because a purely correlational head-importance study would not surface this kind of result.

Spanish-English. We repeat the ablation on 100 Spanish-English images. Spanish and English share Latin script, so ablating script-selective heads cannot change the script of the output. We instead score a *language* flip. We run a language identifier (`langid`, es vs. en) on the generated text and check whether its dominant language changes from the un-ablated baseline. Ablating the top language-selective heads flips the output language above the random-head control in every model (Table 4). The causal relevance of language-selective heads is therefore not specific to Hindi.

Two things differ from Hindi-English. First, the es-head and en-head asymmetry is much smaller, because ablating either language’s heads disrupts the output language by a similar amount. This fits Spanish and English being closer than Hindi and English, since they share both script and much European vocabulary and so are less cleanly separable.

Second, `langid` is noisy on short code-switched generations, which raises the random-head control (up to 33% for InternVL3-2B). We therefore read these results as effects relative to each model’s own control, not as absolute rates.

7 Discussion

Comparing four models rather than two sharpens the picture, and the four play distinct roles in it. Script identity is available early in every vision encoder, but with very different fidelity, and a strong encoder signal is not sufficient for unified routing.

InternVL3-2B. It has the strongest encoder script signal (0.990) and is the one model with near-unified text-image routing, though only for Hindi. This near-merge is not explained by an unusually script-aware encoder, since PaliGemma-3B matches that script-awareness without merging.

PaliGemma-3B. It is the counterexample. Its encoder separates script almost as well as InternVL3-2B’s (0.982 vs. 0.990 peak), yet its routing does not unify at all. Its within-model text-image overlaps sit at the chance level its small head grid dictates (Table 2). A strong vision-encoder script signal is therefore *not* sufficient for unified routing, so routing unification is a property of the decoder side, not the encoder.

Qwen2-VL-2B. Its encoder signal is weaker (0.811) and it keeps the text and image pathways separate. With InternVL3-2B it forms the same-lineage pair that shares 18 of 20 top head coordinates, the maximum in the roster and the anchor of the lineage result below.

SmolVLM. It has the weakest encoder by a wide margin (0.630, not improving with depth), yet it still keeps the pathways separate and its ablation is causal. Its mean-versus-zero reversal is a caution. Causal

ablation without a mean-ablation control can produce clean-looking “language-selective” effects that a fuller control shows to be more complicated.

Lineage over competence. A cross-model pattern concerns *which* heads a model treats as language-important. It tracks decoder lineage in a graded way, from 18 of 20 for the same-lineage pair down to 8–9, 1–2, and zero as lineage distance grows. This pattern rests on a single same-lineage pair, so we hold it tentatively. The ordering holds even in English, where competence is matched, which rules out competence. Architecture still covaries with lineage (Section 8), so we report decoder lineage as the best-supported explanation, not an established cause.

The practical caution holds regardless. Interpretability claims about “the language heads” of one VLM should not be assumed to transfer to a VLM built on the same recipe but a different base decoder, even at similar downstream performance.

8 Limitations

Model scale. All four models are in the 1.7–3B range, a compute constraint rather than a scientific choice. Whether the overlap and lineage effects hold at 7–8B scale is untested.

Covarying architecture. Lineage covaries with head-grid shape, head count, attention scheme, and image compression, none of which we vary independently. We therefore report decoder lineage as the best-supported explanation among the factors we can rule in or out, not an isolated cause. PaliGemma’s small 18×8 grid also raises its chance overlap (Table 2), so its intermediate sharing should be read with that in mind.

Run-to-run stability. The ablation flip rates are stable across three seeds (Appendix C). The head-overlap values are single-run, controlled only by the permutation baseline.

Two language pairs. With only Hindi-English and Spanish-English, we cannot separate script distance from language-family distance as the driver of InternVL3-2B’s Hindi-specific unification. The corpus is also imbalanced (806 vs. 127), though LAHIS uses a matched per-language sample, so this affects only the diversity of the Spanish-English ablation stimuli.

9 Conclusion

We compared the same code-switched content as text tokens and as a rendered image, at head-level granularity, across four vision-language models spanning three decoder lineages. VLMs handle visual and textual language routing differently from one another. Two keep the pathways substantially separate, one nearly unifies them for a single tested language, and one shows no strong pattern either way. *Which heads* a model treats as language-important tracks which decoder it was built on more closely than its vision encoder or its measured language competence, and it falls off in a graded way as lineage distance grows.

The pathway-separation finding is causally confirmed by ablation. The lineage pattern is observational. As discussed in Section 6.3 and the Limitations, it is not fully separable from other architectural differences that covary with lineage across our four models.

Even so, it motivates a practical caution. Head-level interpretability claims about one VLM should not be assumed to transfer to another VLM built on a different base decoder, regardless of how similarly the two perform.

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A Full Script-Probe Trajectories

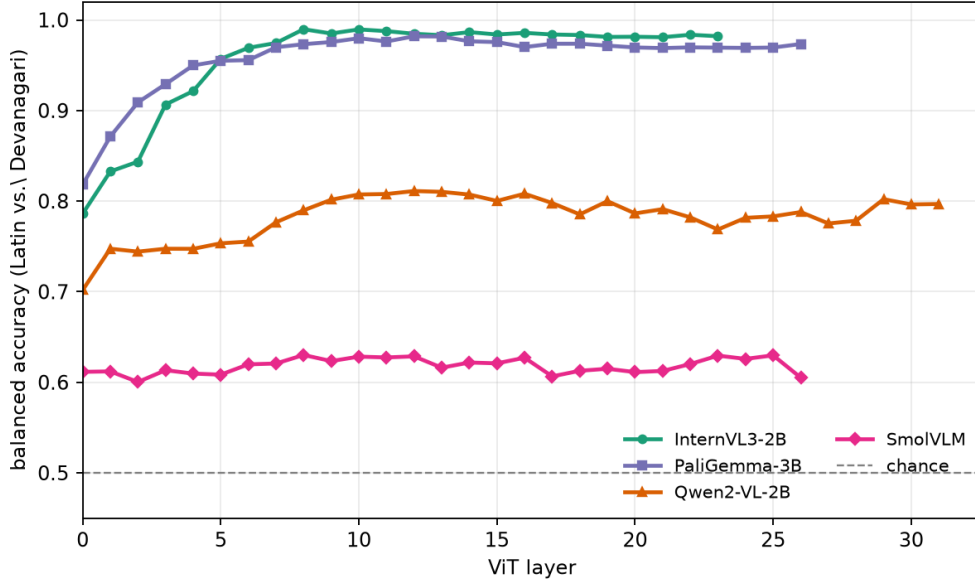


Figure 2: Script-probe balanced accuracy (Latin vs. Devanagari patches, 5-fold CV) at every vision-encoder layer, all four models (dashed line: chance, 0.5). InternVL3-2B and PaliGemma-3B both rise quickly and saturate near 1.0 (0.990 and 0.982 peak). Qwen2-VL-2B rises more slowly to a lower plateau (0.811). SmolVLM stays close to its layer-0 value throughout (0.630 peak), confirming its weak script separability is not a matter of needing more depth. PaliGemma and SmolVLM share the SigLIP-SO400M encoder family yet sit at opposite extremes, a resolution/token-budget effect (Section 6). Section 6 reports the summary numbers (layer 0, peak, final layer) from these curves.

B K -Sweep

Table 5 shows $A \cap C$ and $B \cap C$ at the two extremes of our $K \in \{5, 10, 20, 40, 80\}$ sweep, bracketing the $K=20$ headline in Table 2. The full sweep (all five K values, all four models, all three languages) is released with the code as `checkpoints/head_overlap_matrix.json`.

Table 5: Text-image ($A \cap C$) vs. image-image ($B \cap C$) Jaccard overlap at $K=5$ and $K=80$, the two extremes of the sweep. Qwen2-VL-2B and SmolVLM keep $A \cap C$ well below $B \cap C$ at both extremes, for every language. InternVL3-2B’s Hindi row keeps $A \cap C$ and $B \cap C$ close together at both extremes, the same near-unification Table 2 reports at $K=20$. PaliGemma-3B’s overlaps sit uniformly higher ($A \cap C$ already 0.25 at $K=5$) because its small 18×8 head grid inflates chance-level Jaccard (Table 2); its $A \cap C$ and $B \cap C$ track each other loosely at both extremes, consistent with the flat, chance-dominated within-model structure reported at $K=20$.

Model	Lang	$A \cap C@5$	$A \cap C@80$	$B \cap C@5$	$B \cap C@80$
Qwen2-VL-2B	en	0.000	0.240	0.429	0.539
	es	0.000	0.176	0.429	0.553
	hi	0.111	0.240	0.667	0.650
InternVL3-2B	en	0.000	0.301	0.111	0.468
	es	0.000	0.301	0.000	0.416
	hi	1.000	0.584	1.000	0.739
SmolVLM	en	0.000	0.231	1.000	0.861
	es	0.111	0.203	1.000	0.839
	hi	0.000	0.368	1.000	0.882
PaliGemma-3B	en	0.250	0.495	0.429	0.778
	es	0.250	0.538	0.111	0.616
	hi	0.250	0.468	0.250	0.667

C Seed Variance

To check that the causal-ablation flip rates are stable and not an artifact of one stimulus sample, we repeat the headline Hindi-English ablation under three seeds. Each seed draws an independent random 100-sentence set, and we report the mean and standard deviation of each flip rate across the three seeds in Table 6. The standard deviations are small (at most 8.8 points, and below 5 for most cells). The random-head control stays low under every seed for the three causal-decoder models, and the direction of every zero-ablation effect is preserved. PaliGemma-3B’s Hindi effect is the most stable in the roster at 97.3 ± 0.5 .

Table 6: Hindi-English ablation flip rates (percent) as mean $\pm$ standard deviation across three seeds, each with an independently resampled 100-sentence set. hi-/en-zero and hi-/en-mean zero- and mean-ablate the top-20 script-selective heads; random ablates a matched random head set.

Model	hi-zero	en-zero	random	hi-mean	en-mean
Qwen2-VL-2B	70.3 ± 5.0	46.3 ± 3.4	7.0 ± 4.9	30.3 ± 5.4	33.7 ± 8.8
InternVL3-2B	11.7 ± 1.2	17.0 ± 2.2	14.0 ± 6.2	22.7 ± 3.3	23.3 ± 2.5
SmolVLM	35.3 ± 2.1	35.3 ± 2.1	8.7 ± 0.9	23.3 ± 3.3	47.0 ± 1.6
PaliGemma-3B	97.3 ± 0.5	15.3 ± 2.9	7.0 ± 2.8	29.3 ± 2.6	65.0 ± 2.9

D LAHIS Head-Importance Maps

Figure 3 shows the raw per-head LAHIS importance behind the overlap analysis, for Hindi, as a layer $\times$ head grid per model and condition. The qualitative routing story is visible directly. Qwen2-VL-2B’s text importance (A) is confined to layers 0–1, while its image conditions (B, C) recruit heads throughout the depth, so the top-20 sets barely intersect, matching the separated pathways of Table 2.

InternVL3-2B’s A, B and C light up nearly the same early heads, its Hindi near-unification. SmolVLM’s importance is sparse and scattered across its wide 24×32 grid. PaliGemma-3B’s importance is spread diffusely across layers with no sharp concentration, so its top-20 sets overlap only at the chance level its small 18×8 grid dictates.

LAHIS per-head importance (Hindi), top-20 heads ringed

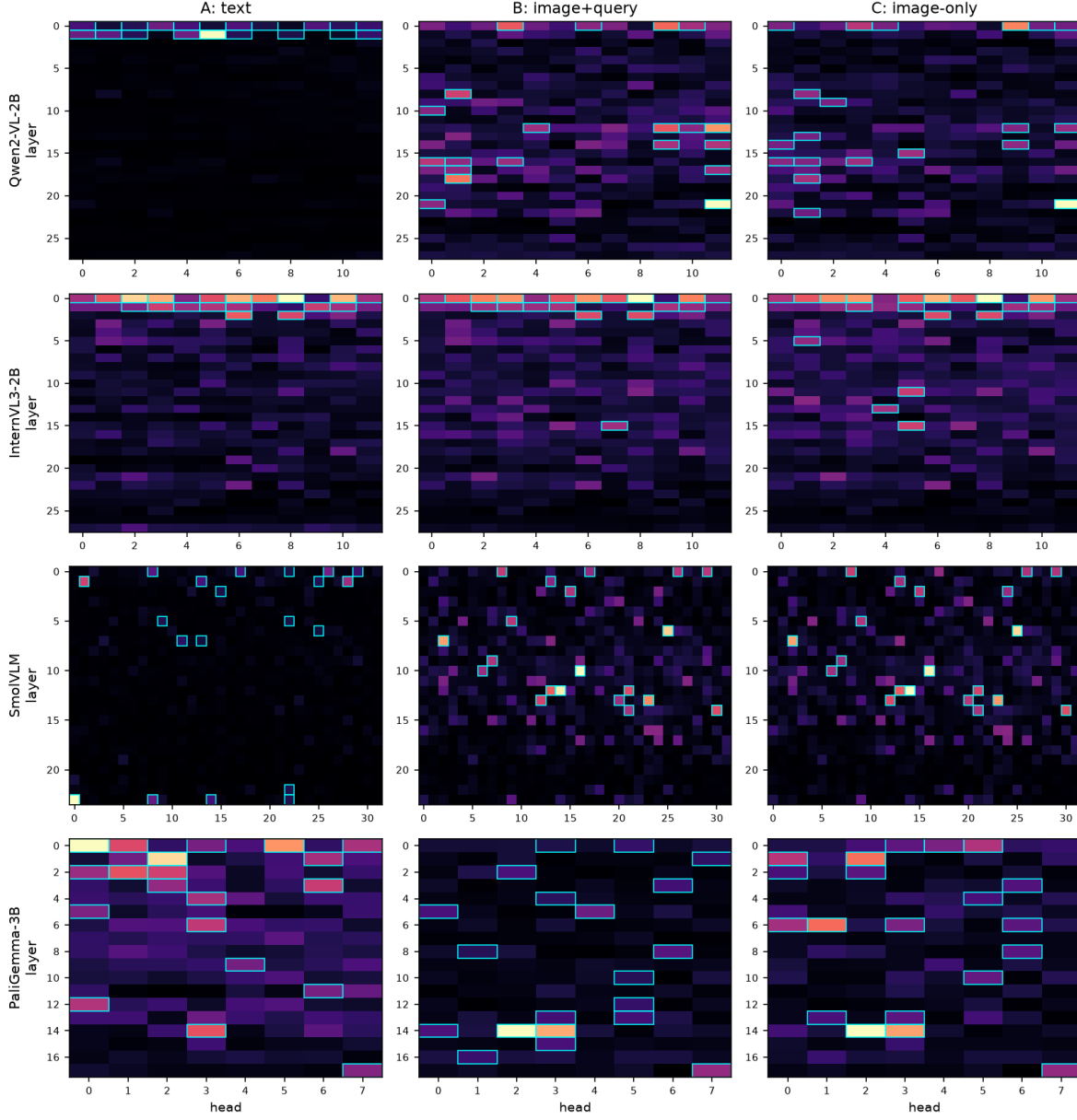


Figure 3: LAHIS per-head importance for Hindi, min-max normalized within each panel. Rows are the four models and columns the three routing conditions (A=text, B=image+query, C=image-only). Cyan rings mark each panel’s top-20 heads (the set entering the overlap analysis). Grid shapes differ by model (28×12, 28×12, 24×32, 18×8). Brighter is more important.

LAHIS per-head importance (Spanish), top-20 heads ringed

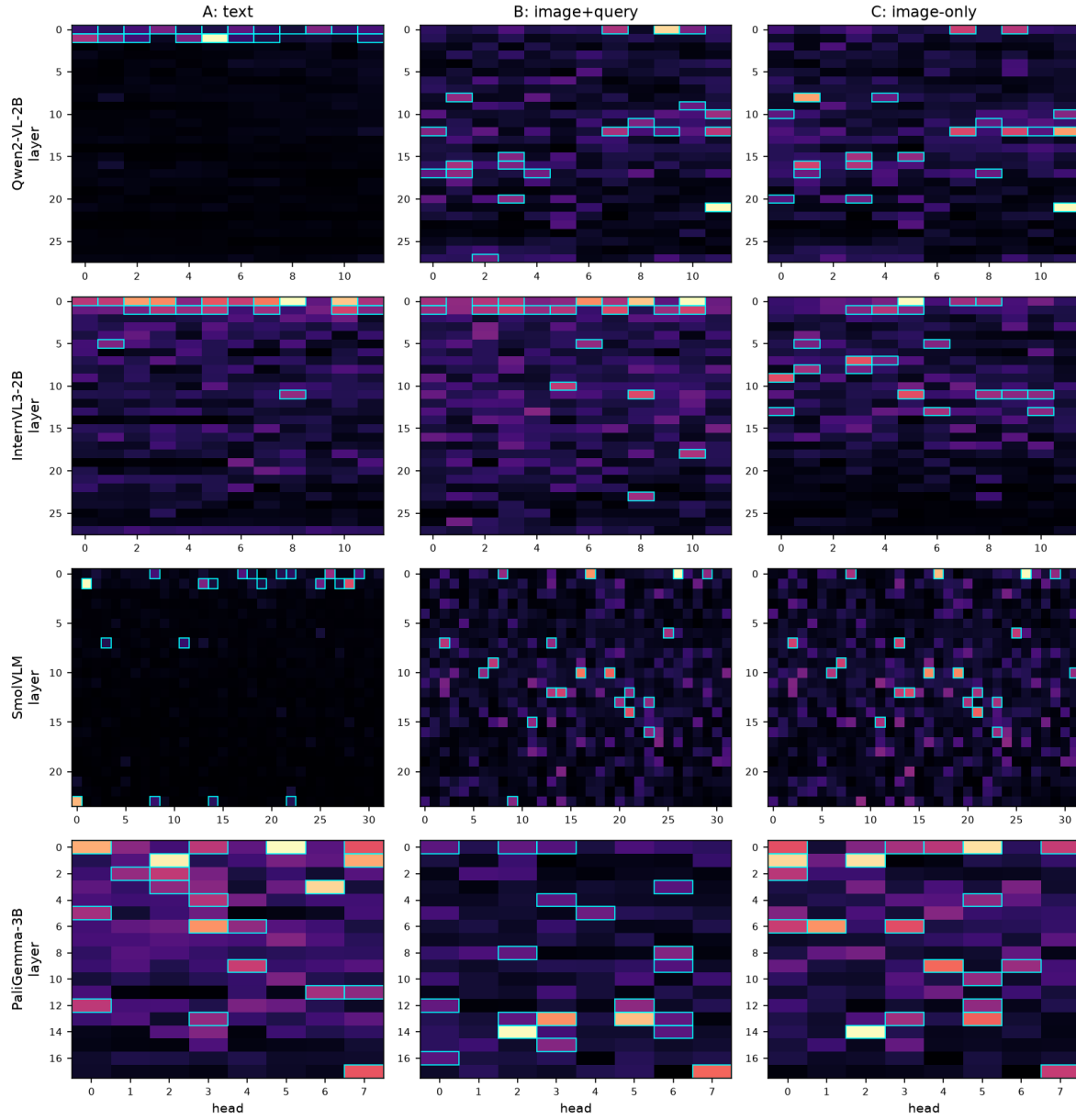


Figure 4: LAHIS per-head importance for Spanish, plotted as in Figure 3 (rows are the four models, columns the three routing conditions, cyan rings mark each panel’s top-20 heads). The load-bearing head sets sit in the same layer bands as for Hindi, but for InternVL3-2B the text and image panels align less closely than they do for Hindi, consistent with its lower Spanish text-image overlap in Table 2.

E Sample Stimuli

Figure 5 shows representative rendered code-switched stimuli. Hindi-English items mix Devanagari and Latin script within a sentence; Spanish-English items stay in Latin script throughout, which is why the causal script-flip metric (Section 5) applies only to the Hindi-English pair.

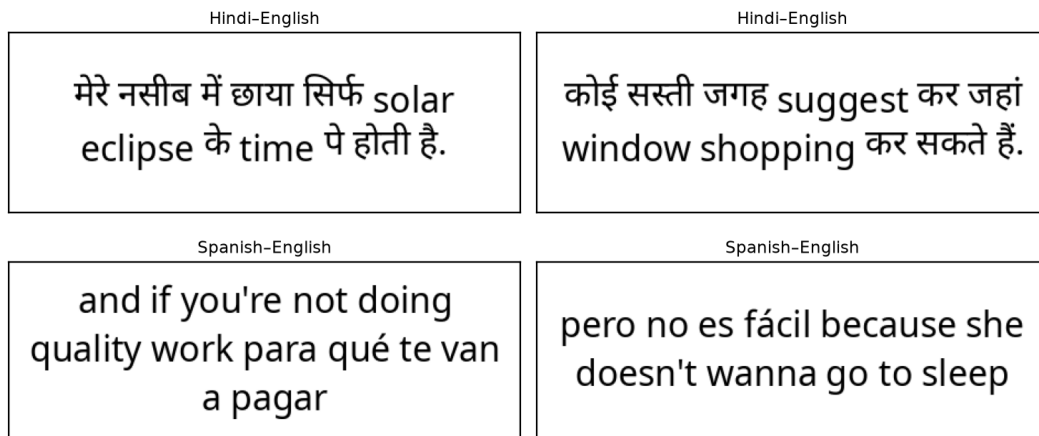


Figure 5: Four rendered code-switched sentences from the evaluation set (two Hindi-English, two Spanish-English), shown as the models see them, rendered to an image with no accompanying prompt tokens.